

Digestive Physiology & Digestive Conditions

Today's Materials

- Test tubes
- Test tube racks
- Starch Solution
- Hard Boiled Eggs
- Stock solutions of amylase, pepsin, and trypsin
- pH buffer solutions of 7.0, 5.0, and 2.0
- Hot water bath
- Dropper bottles of Benedict's reagent
- Disposable transfer pipettes
- Wax pencils for writing on test tubes
- Liquid disposal jug
- Blank Pages in Lab Notebook
- Extra Textbooks/Manuals in Room

Notes:

- The various solutions/materials for today's lab have been spread around to different areas of the room. This is to try to prevent cross contamination of the solutions. It's harder to mix solutions up when they are farther apart! Do not use the same pipette or the same measuring equipment with different solutions.
- Wear gloves throughout the experiments and dispose of test tube contents in the designated jug.

Activity 1: Observing Chemical Digestion

In this activity, your team is challenged to simulate the chemical digestive activities of 3 digestive system regions using test tubes.

1. Pick up six test tubes and number them 1s, 1e, 2s, 2e, 3s, and 3e. Additionally mark each tube with a symbol or letters to identify them as your team's test tubes. Place them in a tube rack on your table.
2. Tubes with a 1 will represent the oral cavity. Tubes with a 2 will represent the stomach. Tubes with a 3 will represent the duodenum (as indicated in Table 1).
3. Use background knowledge, information from your textbook, or other information you can look up to complete the first two blank columns of Table 1.
4. Add 3 ml of starch solution to each tube with an "s" in its label.
5. Add a small, smooth slice of egg white to each tube with an "e" in its label.
6. Fill out the *Which enzymes? Bile salts?* column of Table 1 to indicate which enzymes you think you would find in each location.
7. Fill out the *Target pH* column of Table 1 to indicate what the pH might typically be inside each organ.
8. *Check your ideas with your instructor!*
9. Add 3 ml of each indicated enzyme into the appropriate tubes.

10. Add 5 ml of the appropriate acid/base buffer solution to each test tube out of recognition for the pH in that location (note that this might not dramatically impact the pH of the solution as a whole, but it will at least make it closer to real life).
11. Gently mix the contents of the test tubes and place them in the water bath. Begin timing so you know how long the test tubes have been in the water bath.
12. Complete the *Tests to Perform* and *Expected Results* columns of Table 1.
13. *Check your ideas with your instructor!*
14. **While you are waiting 60 minutes, complete the Activity 2 observations below!**
15. After 60 minutes, closely observe the “e” test tubes that have egg in them. Note whether the egg white appears to have broken down at all. The egg white might appear smaller, with edges eaten away, and the solution might appear cloudy. These signs would be a positive result for protein digestion. Record your results under *Actual Test Results* in the table.
16. Next, test for sugars in the “s” test tubes with starch in them. To do this, add some drops of Benedict’s reagent to the test tube (up to a few ml). If the solution...
 - Appears blue, like the Benedict’s starts, there is no sugar
 - Appears greenish, there is a small amount of sugar
 - Appears yellowish, there is a moderate amount of sugar
 - Appears orange or reddish, there is a large amount of sugarRecord your results under *Actual Test Results* in the table.
17. To what extent did your results match your predictions? What might be the causes of any discrepancies?
18. Describe how well your test tubes simulated the oral cavity, stomach, and duodenum. In what ways were our test tubes accurate representations vs. not?

Activity 2: Digestive Organs of Chemical Digestion and Nutrient Storage

The activities below are an extension of the work we completed in last week’s lab, so you will notice that the format is similar to last week’s activities. The organs we’re studying today are more deeply connected with chemical digestion or nutrient storage.

You may complete these activities in any order! For example, you could begin with the pancreas activities instead of the small intestine activities. If the materials for an activity are all being used by others, just skip to a different activity and come back to the others!

Small Intestine

Small Intestine Models and/or Diagrams

Small Intestine Microscope Slide

Small Intestine Specimen(s) and/or Medical Images

Small Intestine Information Sheets

1. Write “Small Intestine” on the diagram from *last week’s* lab to indicate where you see this organ.
2. In your notes, create a rough sketch of the small intestine. Label the duodenum, jejunum, and ileum.
3. What do you notice about the small intestine specimen(s) or medical images in the room? Which features of these organs or structures you labeled above can you see?
4. Set up a microscope with a view of the small intestine slide and record a diagram of your microscope view in your notebook. Look up a diagram of small intestine histology either in a book or online. Label a villus and the simple columnar epithelium.
5. Review the provided information regarding gastroenteritis. Return to the diagrams you made above. Indicate through labels or additional diagramming the region(s) of the intestine likely most related to lactose intolerance following gastroenteritis, as well as the areas of your microscope slide view most related to this phenomenon. Why did you choose those particular regions?

Liver & Gall Bladder

Liver/Gall Bladder Models and/or Diagrams

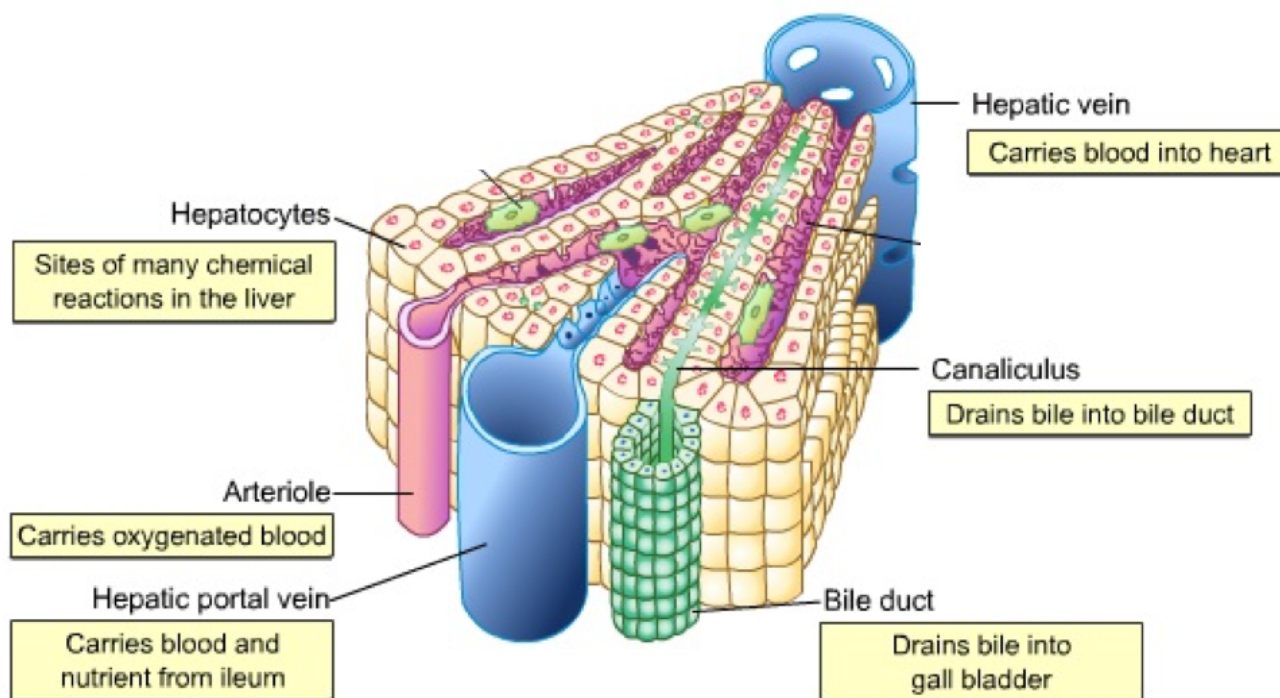
Liver Lobules Microscope Slide

Liver Specimen(s) and/or Medical Images

Liver/Gall Bladder Information Sheets

1. Write “Liver” on the diagram from *last week’s* lab to indicate where you see this organ.
2. In your notes, create a rough sketch of the liver and gall bladder. Label the left and right lobes of the liver, the falciform ligament, the gall bladder, and the bile duct.
3. Which of these structures connects the liver to the diaphragm and abdomen wall?
4. What do you notice about the liver specimen(s) or medical images in the room? Which features of these organs or structures you labeled above can you see?
5. Set up a microscope with a view of the liver slide and record a diagram of your microscope view in your notebook. Look up a diagram of a liver lobule either in a book or online. Label the central/hepatic vein, hepatocytes, and the location of a portal triad.
6. The diagram on the next page depicts one, small section of the liver lobule you just drew from your microscope view. You can see the central/hepatic vein, hepatocytes, and portal triad (hepatic artery, hepatic portal vein, bile duct). Consult the provided information regarding jaundice and codeine.
7. On the diagram on the next page, use a pen or pencil to write a “B” in the location where bile is produced. Next, draw one or more arrow(s) to show the path bile would normally take to leave the liver and enter the gall bladder. Finally, use a different color of pen or pencil to draw arrows showing the path bile would take to get into the blood if the bile duct is blocked.
8. Also on the diagram on the next page, write a “C” in to indicate the vessel codeine would use to enter the liver from the intestine. Also, write an “M” in the

location where the codeine would be converted to morphine. Finally, draw one or more arrows to show the path the codeine (and later morphine) would take to get into wider circulation around the body.



Pancreas

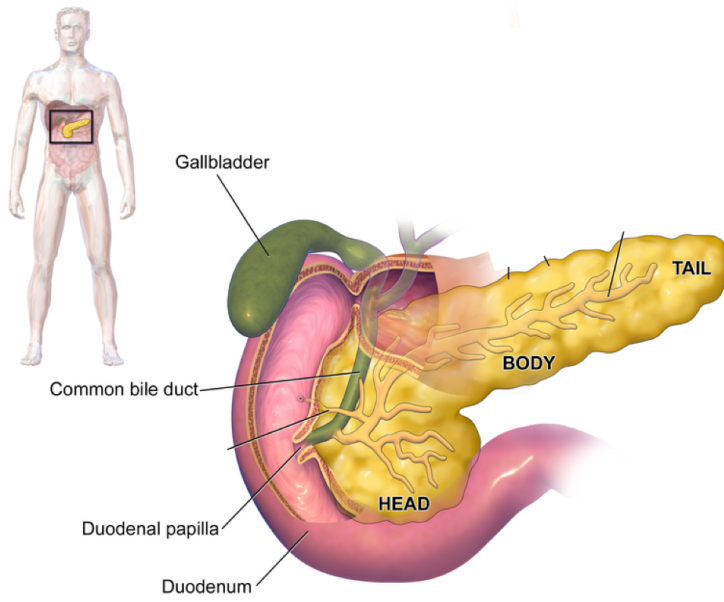
Pancreas Models and/or Diagrams

Pancreas Microscope Slide

Pancreas Specimen(s) and/or Medical Images

Pancreas Information Sheets

1. Write "Pancreas" on the diagram from *last week's* lab to indicate where you see this organ.
2. In your notes, create a rough sketch of the pancreas. Label the lobules and pancreatic duct.
3. What do you notice about the pancreas specimen(s) or medical images? Which features of the pancreas or structures you labeled above can you see?
4. Set up a microscope with a view of the pancreas slide and record a diagram of your microscope view in your notebook. Look up a diagram of pancreas cells either in a book or online. Label the acinar cells and pancreatic duct.
5. The diagram on the next page shows the pancreas's normal orientation compared to other digestive structures.
6. Consult the provided information about pancreatitis. In your notebook, create a rough diagram similar to the one below, but showing pancreatitis and specifically how it might cause jaundice.



Reviewing Some More Digestive Organ Fun Facts

Record the answers to the following:

1. Which of the organs studied today is the heaviest internal organ in the body?
2. Which of the organs studied today was mistakenly believed to hold air and assist with breathing?
3. Which of the organs studied today was originally thought to be nothing more than a shock absorber in the gut.

<i>Table 1</i>	Background Information		Preparing Test Tubes			Gathering Results (after 1 hour)		
Location	Molecule Type(s) Digested Here	Resulting Molecule(s) After Digestion	Nutrient Sources	Which enzymes?	Target pH	Test(s) to Perform After Digestion	Expected Test Results	Actual Test Results
Oral Cavity (Tube 1s)			Starch Solution			Benedict's		
Oral Cavity (Tube 1e)			Egg White			Observation for Protein Digestion		
Stomach (Tube 2s)			Starch Solution			Benedict's		
Stomach (Tube 2e)			Egg White			Observation for Protein Digestion		
Duodenum (Tube 3s)			Starch Solution			Benedict's		
Duodenum (Tube 3e)			Egg White			Observation for Protein Digestion		